FoSCoS — Officer Portal (FSO) Specification

Scope: A separate web application for a single Food Safety Officer (FSO). It reads from and writes to the same Firebase database as the citizen-facing FoSCoS app. All actions taken here must reflect in real time on the FBO's dashboard.

Authentication

Login Screen

- Email + password login (no OTP — this is a credential-based internal portal)
- No self-registration; accounts are provisioned by an admin
- On successful login, land on the Dashboard page
- Session persists until explicit logout

Navigation Structure

Persistent left sidebar with the following pages:

Nav Item	Icon	Description
Dashboard	📊	Overview stats and charts
Applications	📋	All pending permanent license applications
Temp Licenses	⚡	Temp license monitoring and upgrade tracking
FBO Directory	🏪	All registered businesses

Inspections	🔍	Scheduled and completed inspections
Grievances	📣	Consumer complaints linked to FBOs
Settings	⚙️	Officer profile

Active page is highlighted. Officer name and logout appear at the bottom of the sidebar.

Page 1 — Dashboard

Purpose: At-a-glance snapshot of this month's activity.

Stats Row (top)

Four number cards, each showing current month value + change vs. last month:

- Total Applications Received
- Applications Approved
- Applications Rejected / Returned
- Inspections Completed

Charts Section

Two charts side by side:

Chart 1 — Applications by Status (Pie/Donut) Segments: Pending Review, Documents Requested, Inspection Scheduled, Approved, Rejected. Values pulled live from Firebase.

Chart 2 — Applications Over Time (Bar Chart) X-axis: last 6 months. Y-axis: count. Two bars per month — Received vs. Approved. Lets the officer see processing backlog trend.

Alerts Panel

A scrollable list of items that need attention today:

- Applications older than 30 days with no action taken

- Temp licenses expiring within 7 days with no upgrade started
- Inspections scheduled for today
- Grievances unresolved for more than 14 days

Each alert is a clickable row that navigates directly to the relevant record.

Recent Activity Feed

Last 10 actions taken by the officer (approvals, rejections, flags, inspections scheduled), with timestamp. Read-only audit trail of their own actions.

Page 2 — Applications

Purpose: The primary work queue. All permanent license applications submitted by FBOs.

Filters Bar

- Search by FBO name or Application ID
- Filter by Status: All / Pending / Documents Requested / Inspection Scheduled / Approved / Rejected
- Filter by License Category: Basic Registration / State License / Central License
- Sort by: Date Filed (default) / Last Updated / Days Pending

Applications Table

Columns: Application ID · FBO Name · License Category · Date Filed · Days Pending · Current Status · Action button

Rows are color-coded by urgency:

- Red background: pending for more than 30 days
- Amber background: documents requested but not resubmitted for more than 7 days
- White: normal

Clicking any row opens the Application Detail page.

Application Pipeline

Each application moves through 5 stages. The current stage is shown as a horizontal step indicator on the detail page:

1. Submitted → 2. Documents Review → 3. Inspection → 4. Final Review → 5. Approved

Not all applications require Stage 3 (Inspection is mandatory for State and Central licenses; optional for Basic Registration at the officer's discretion).

Page 3 — Application Detail

Opened by clicking any row in the Applications table. This is where most of the officer's real work happens.

Header

- FBO name, Application ID, license category, date filed
- Current pipeline stage (step indicator)
- Status badge
- Action buttons row (see Actions section below)

Tabs

Tab 1 — Business Info Read-only display of everything the FBO submitted: business name, owner name, address, KOB category, turnover, food items, ID type and number, contact details. Data pulled directly from Firebase.

Tab 2 — Documents List of all documents the FBO uploaded, with:

- Document name
- Upload date
- Preview button (opens file from Firebase Storage URL in a new tab)
- Status badge: Verified / Pending / Returned

- Per-document action buttons: **Approve** or **Return**

When the officer clicks **Return** on a specific document:

- A text input appears for the officer to type the reason
- On confirm, that document's status in Firebase updates to returned
- The FBO's dashboard shows the document as returned with the reason text visible

The "Next" button on this tab only activates when all documents are either Verified or explicitly flagged as missing (officer acknowledges the gap).

Tab 3 — Audit Trail Chronological log of every action taken on this application by anyone (officer or system). Each entry shows: action taken, who did it, timestamp, and any notes attached. This is append-only — nothing can be deleted.

Tab 4 — Flag History Shows any flags that have been raised on this application, when, and the reason. If no flags, shows a clean state.

Actions (Header Button Row)

Advance to Next Step Moves the application to the next pipeline stage. Opens a modal where the officer can optionally add a note. The note is visible in the audit trail but not shown to the FBO.

Request Documents Opens a modal with:

- Checklist of all documents for this license category (auto-populated based on category)
- Officer checks which ones are missing or need replacement
- A text field for additional instructions to the FBO
- On confirm: application status changes to documents_requested. FBO receives an SMS/notification. On their dashboard, a banner appears listing the requested documents with the officer's instructions. The FBO can re-upload and resubmit.

Schedule Inspection *(see Inspections page for full detail)* Opens inspection scheduling modal. On confirm, creates an inspection record linked to this application.

Approve Application Final approval action. Only available when the application is at Stage 4 (Final Review) and all documents are verified.

- Opens a confirmation modal
- On confirm: application status changes to approved in Firebase, a license number is generated and written to the FBO's record, the FBO's dashboard switches to the "Active License" state and the license card appears with the issued details

Reject Application Available at any stage.

- Opens a modal requiring the officer to:
    1. Select a rejection reason from a dropdown (document fraud / ineligible business type / incomplete information / failed inspection / other)
    2. Write a detailed report (minimum 50 characters, enforced)
    3. Set a resubmission window: dropdown of 7 / 14 / 30 / 60 days
- On confirm: application status changes to rejected. On the FBO's dashboard, a rejection banner appears with the reason, the full report text, and a countdown showing days remaining to resubmit. A "Modify and Resubmit" button appears on the FBO side, which reopens their application in edit mode. On resubmission, the application re-enters the queue at Stage 1 and the officer sees it flagged as a resubmission.

Flag Application Raises an internal flag visible only to officers. Does not notify the FBO. Used for: suspicious activity (duplicate address, mismatched IDs), escalation notes, or watchlist markers. Flagged applications show a 🚩 indicator in the applications table.

Page 4 — Temp Licenses

Purpose: Dedicated monitoring page for all Instant Temporary Licenses issued through the citizen portal. The officer does not approve or reject temp licenses (they are issued automatically) but monitors compliance and upgrade activity.

Summary Cards Row

- Total Active Temp Licenses
- Expiring Within 7 Days (urgent, shown in red)
- Overdue — Expired Without Upgrade (shown in red)
- Upgrades Completed This Month

Temp License Table

Columns: License Number · FBO Name · Setup Type · Issued Date · Expiry Date · Days Remaining · FoSTaC Done · Water Test Done · Status

Status values:

- Active — within validity, no action needed
- Expiring Soon — less than 7 days remaining
- Overdue — past expiry, no upgrade application filed
- Upgraded — FBO has filed a permanent license application

Rows sortable by Days Remaining (default, ascending so most urgent is at top).

Actions Per Row

- **View FBO** — opens their record in FBO Directory
- **Send Reminder** — triggers an automated SMS to the FBO reminding them to upgrade (records the action in audit trail)
- **Schedule Inspection** — for overdue cases, the officer can schedule a compliance check visit
- **Mark as Cancelled** — if the business is known to have closed, removes it from the active monitoring list with a reason note

Upgrade Funnel Chart

A simple funnel or bar chart showing: Temp Licenses Issued (last 90 days) → FoSTaC Enrolled → Water Test Done → Permanent Application Filed → Permanent License Granted. Gives the officer a sense of how many FBOs are progressing through the compliance path.

Page 5 — FBO Directory

Purpose: Full registry of all businesses that have ever interacted with FoSCoS — licensed, pending, rejected, or temp-only.

Filters

- Search by name, license number, or owner name
- Filter by License Type: Basic / State / Central / Temp Only
- Filter by Compliance Status: Compliant / Expiring Soon / Overdue / Suspended
- Filter by area/district

Directory Table

Columns: FSSAI License No. · Business Name · Category · Location · License Expiry · Compliance Status · Last Inspection

Clicking a row opens the FBO Profile.

FBO Profile

A full view of a single business across tabs:

Business Details tab — All registered information. Read-only.

License History tab — All licenses issued to this FBO over time, with issue and expiry dates.

Inspection History tab — All past inspections: date, officer, outcome (Pass / Advisory / Fail), notes.

Applications tab — All applications ever filed, with their current status. Each links to the Application Detail page.

Grievances tab — Consumer complaints submitted about this business (linked from the Grievances page).

Page 6 — Inspections

Purpose: Manage all inspection scheduling and outcomes.

Scheduling an Inspection

Triggered either from an Application Detail page or from the Temp License table. The scheduling modal collects:

- FBO name (auto-filled from context)
- Inspection type: Routine / Pre-License / Complaint-Based / Compliance Check (Temp)
- Date (date picker — no past dates)
- Time (time picker)
- Notes for context

On confirm, a record is written to Firebase. The FBO's Application Tracker page shows a new "Inspection Scheduled" step with: inspection type, date, time, and the officer's **last name only** (e.g. "Officer Kumar"). No contact number shown to FBO.

Inspections Table

Columns: FBO Name · Type · Scheduled Date · Status · Outcome

Status: Scheduled / Completed / Cancelled / Missed

Recording an Outcome

After an inspection is marked Complete, the officer fills a short outcome form:

- Overall result: Pass / Advisory / Fail
- Notes (free text)
- Checklist of items inspected (kitchen hygiene, labelling, storage, equipment, food item compliance) — each marked Pass / Fail
- Option to upload photos

On save, the outcome is written to Firebase. The FBO's Inspection tab on their dashboard updates to show the result. If the result is Fail, the application is automatically moved back to Stage 2 (Documents Review) and the officer is prompted to send a Request Documents action.

Page 7 — Grievances

Purpose: Consumer complaints submitted through the Consumer Corner on the citizen portal, linked to specific FBOs.

Grievances Table

Columns: Complaint ID · FBO Name · Category (food quality / unlicensed / hygiene / labelling / other) · Filed Date · Days Open · Status

Status: Open / Under Review / Resolved / Escalated

Grievance Detail

Clicking a row shows:

- Full complaint text submitted by the consumer
- FBO details
- Action buttons:
    - **Mark Under Review** — changes status, no FBO notification
    - **Trigger Inspection** — creates an inspection linked to both the grievance and the FBO record
    - **Resolve** — requires the officer to write a resolution note; changes status to Resolved; FBO is not notified but the resolution is logged
    - **Escalate** — marks for higher-level attention (currently just a flag; no routing in v1)

Page 8 — Settings

Officer profile page. Displays:

- Full name, email, officer ID, designation, assigned region
- Edit profile (name, contact number only — email and officer ID are read-only)
- Change password
- Notification preferences: toggle SMS alerts on/off for: new applications assigned, grievances filed, inspections due today
- Logout button

Firebase Data Contracts

What the officer portal reads from Firebase:

- applications/{id} — full application document including all FBO-entered fields, document URLs, and current status
- users/{userId} — FBO account details
- tempLicenses/{id} — all temp license records
- inspections/{id} — all inspection records
- grievances/{id} — all consumer complaints

What the officer portal writes to Firebase:

- applications/{id}.status — updated on every action
- applications/{id}.auditTrail — append-only array; every action pushes a new entry with {action, officerId, timestamp, note}
- applications/{id}.rejectionReport — written on rejection: {reason, reportText, resubmitWindowDays, rejectedAt}
- applications/{id}.documentsRequested — array of document names with the officer's instructions
- applications/{id}.currentStep — integer 1–5
- applications/{id}.licenseNumber — written on approval
- inspections/{id} — created on schedule, updated on outcome
- tempLicenses/{id}.reminderSentAt — timestamp when officer sends reminder
- tempLicenses/{id}.status — can be changed to cancelled
- grievances/{id}.status — updated on each action
- grievances/{id}.resolutionNote — written on resolve

FBO dashboard reactivity:

The citizen app listens to applications/{id} in real time using Firebase onSnapshot. Any write by the officer portal is immediately reflected on the FBO's screen without a page reload. This covers: status changes, document return reasons, rejection reports, and inspection scheduling.

Application Status State Machine

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submitted
  → documents_requested   (officer requests docs)
    → submitted          (FBO resubmits)
  → inspection_scheduled  (officer schedules)
    → inspection_complete (outcome recorded)
  → final_review          (officer advances manually)
  → approved              (final approval)
  → rejected              (at any stage; FBO can resubmit within window)
```

Temp license statuses are separate and managed independently on the tempLicenses collection.

Notifications Sent to FBO (triggered by officer actions)

Officer Action	FBO Notification
Request Documents	SMS: "FSSAI officer has requested additional documents for your application [ID]. Login to FoSCoS to view details."
Schedule Inspection	SMS: "An FSSAI inspection has been scheduled on [date] at [time]. Officer: [LastName]."
Approve Application	SMS: "Congratulations! Your FSSAI license [number] has been approved. Login to download your certificate."
Reject Application	SMS: "Your FSSAI application [ID] has been rejected. You have [X] days to modify and resubmit. Login to view the officer's report."
Return Document	In-app only: document row shows returned status + reason

Send Temp Reminder	SMS: "Your Temporary FSSAI License expires in [X] days. Please apply for a permanent license to continue operations."

All SMS messages are dispatched via MSG91 (same provider as the citizen portal OTP). The officer portal does not send emails — only SMS and in-app updates.

Version: v1.0 — Single FSO, single jurisdiction. Multi-officer hierarchy and role-based access deferred to v2.